

- [14] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, “Dropout: a simple way to prevent neural networks from overfitting,” *Journal of Machine Learning Research*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [15] X. Liu, M. Cheng, H. Zhang, and C.-J. Hsieh, “Towards robust neural networks via random self-ensemble,” in *European Conference on Computer Vision*, 2018, pp. 369–385.
- [16] H. Inayoshi and T. Kurita, “Improved generalization by adding both auto-association and hidden-layer-noise to neural-network-based-classifiers,” in *IEEE Workshop on Machine Learning for Signal Processing*, 2005.
- [17] C. M. Bishop, “Training with noise is equivalent to tikhonov regularization,” *Neural Computation*, vol. 7, no. 1, pp. 108–116, jan 1995.
- [18] G. An, “The effects of adding noise during backpropagation training on a generalization performance,” *Neural Computation*, vol. 8, no. 3, pp. 643–674, apr 1996.
- [19] K. Audhkhasi, O. Osoba, and B. Kosko, “Noise benefits in backpropagation and deep bidirectional pre-training,” in *International Joint Conference on Neural Networks*, aug 2013.
- [20] A. Neelakantan, L. Vilnis, Q. V. Le, I. Sutskever, L. Kaiser, K. Kurach, and J. Martens, “Adding gradient noise improves learning for very deep networks,” *arXiv preprint arXiv:1511.06807*, 2015.
- [21] A. Graves, “Practical variational inference for neural networks,” in *Advances in Neural Information Processing Systems*, J. Shawe-Taylor, R. S. Zemel, P. L. Bartlett, F. Pereira, and K. Q. Weinberger, Eds., 2011, pp. 2348–2356.
- [22] A. Murray and P. Edwards, “Synaptic weight noise during multilayer perceptron training: fault tolerance and training improvements,” *IEEE Transactions on Neural Networks*, vol. 4, no. 4, pp. 722–725, jul 1993.
- [23] C. Blundell, J. Cornebise, K. Kavukcuoglu, and D. Wierstra, “Weight uncertainty in neural network,” in *International Conference on Machine Learning*, vol. 37, 2015, pp. 1613–1622.
- [24] Y. Bengio, N. Léonard, and A. Courville, “Estimating or propagating gradients through stochastic neurons for conditional computation,” *arXiv preprint arXiv:1308.3432*, 2013.
- [25] D. J. MacKay, “A practical bayesian framework for backpropagation networks,” *Neural computation*, vol. 4, no. 3, pp. 448–472, 1992.
- [26] A. Graves, “Practical variational inference for neural networks,” in *Advances in neural information processing systems*, 2011, pp. 2348–2356.
- [27] G. Hinton and D. Van Camp, “Keeping neural networks simple by minimizing the description length of the weights,” in *Proc. of the 6th Ann. ACM Conf. on Computational Learning Theory*. Citeseer, 1993.
- [28] K. Shridhar, F. Laumann, and M. Liwicki, “A comprehensive guide to bayesian convolutional neural network with variational inference,” *arXiv preprint arXiv:1901.02731*, 2019.
- [29] A. Kendall and Y. Gal, “What uncertainties do we need in bayesian deep learning for computer vision?” in *Advances in neural information processing systems*, 2017, pp. 5574–5584.
- [30] Y. Kwon, J.-H. Won, B. Kim, and M. Paik, “Uncertainty quantification using bayesian neural networks in classification: Application to ischemic stroke lesion segmentation,” *Computational Statistics and Data Analysis*, 04 2018.
- [31] K. Shridhar, F. Laumann, and M. Liwicki, “Uncertainty estimations by softplus normalization in bayesian convolutional neural networks with variational inference,” *arXiv preprint arXiv:1806.05978*, 2018.
- [32] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *Advances in neural information processing systems*, 2017, pp. 6402–6413.

- [33] H. Wang, X. Shi, and D. Yeung, “Natural-parameter networks: A class of probabilistic neural networks,” *CoRR*, vol. abs/1611.00448, 2016. [Online]. Available: <http://arxiv.org/abs/1611.00448>
- [34] J. Gast and S. Roth, “Lightweight probabilistic deep networks,” *CoRR*, vol. abs/1805.11327, 2018. [Online]. Available: <http://arxiv.org/abs/1805.11327>
- [35] J. M. Hernández-Lobato and R. Adams, “Probabilistic backpropagation for scalable learning of bayesian neural networks,” in *International Conference on Machine Learning*, 2015, pp. 1861–1869.
- [36] A. Krizhevsky, V. Nair, and G. Hinton, “Cifar-10 (canadian institute for advanced research).” [Online]. Available: <http://www.cs.toronto.edu/~kriz/cifar.html>
- [37] A. Coates, A. Ng, and H. Lee, “An analysis of single-layer networks in unsupervised feature learning,” in *International Conference on Artificial Intelligence and Statistics*, 2011, pp. 215–223.
- [38] A. L. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, “Learning word vectors for sentiment analysis,” in *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*. Portland, Oregon, USA: Association for Computational Linguistics, June 2011, pp. 142–150. [Online]. Available: <http://www.aclweb.org/anthology/P11-1015>
- [39] D.-A. Clevert, T. Unterthiner, and S. Hochreiter, “Fast and accurate deep network learning by exponential linear units (elus),” *arXiv preprint arXiv:1511.07289*, 2015.
- [40] G. Klambauer, T. Unterthiner, A. Mayr, and S. Hochreiter, “Self-normalizing neural networks. arxiv 2017,” *arXiv preprint arXiv:1706.02515*, 2017.
- [41] F. Laumann, K. Shridhar, and A. L. Maurin, “Bayesian convolutional neural networks,” *CoRR*, vol. abs/1806.05978, 2018. [Online]. Available: <http://arxiv.org/abs/1806.05978>
- [42] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [43] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov, “Improving neural networks by preventing co-adaptation of feature detectors,” *arXiv preprint arXiv:1207.0580*, 2012.
- [44] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, “Extracting and composing robust features with denoising autoencoders,” in *ACM International Conference on Machine Learning*, 2008, pp. 1096–1103.